

Keagan Ryder

CS 499

Module Six Journal

Part One: Emerging and Disruptive Technologies

Two emerging and disruptive technologies that have fundamentally reshaped computer science are artificial intelligence–driven automation and cloud-native computing platforms.

Artificial intelligence (AI), particularly machine learning and large language models, represents a major shift from rule-based systems to data-driven, adaptive systems capable of learning patterns and making decisions at scale. This technology has changed how software is designed, tested, and deployed, enabling intelligent automation, predictive analytics, and real-time personalization across industries. Cloud-native computing, including containerization and scalable cloud infrastructures, has disrupted traditional on-premises computing by enabling elastic resource allocation, global accessibility, and continuous integration and deployment pipelines.

The impact of these technologies on computer science and my career path is significant. AI-driven systems require strong foundations in algorithms, data structures, and software architecture to ensure correctness, performance, and ethical use. Cloud platforms demand an understanding of distributed systems, APIs, and database integration. As a computer science professional, these technologies directly align with my focus on full-stack development, mobile application design, and scalable system architecture. Experience with AI-assisted development tools and cloud-based deployment environments strengthens my ability to design modern, production-ready software systems.

From a broader human and societal perspective, AI and cloud computing have transformed how people work, communicate, and access services. AI has improved efficiency in healthcare, logistics, and accessibility technologies while also raising ethical concerns around data privacy, bias, and workforce displacement. Cloud computing has enabled remote work, global collaboration, and rapid innovation, lowering the barrier to entry for startups and independent developers. Together, these technologies continue to reshape communities by increasing digital access while also highlighting the need for responsible system design and governance.

Throughout this course, I have achieved several key course outcomes, including the ability to design and evaluate computing solutions, apply appropriate algorithms and data structures, and communicate technical concepts effectively through documentation and professional artifacts. I have also demonstrated the integration of databases into application architectures and reflected on ethical and societal impacts of computing technologies. Remaining outcomes focus primarily on finalizing enhancements, refining documentation, and presenting a polished, cohesive ePortfolio that clearly communicates my technical growth and professional readiness.

Part Two: ePortfolio Artifact Status Update

Category	SDE	Algorithms	Databases
Name	CS 360	CS 300	CS 360
Status of IE	Complete	Complete	Complete
Submission Status	Submitted	Submitted	Submitted
Status of FE	Complete	Complete	In Progress
Upload to portfolio	Submitted	Submitted	In Progress

Category	SDE	Algorithms	Databases
Finalized Portfolio	In Progress	In Progress	In Progress

At this stage, all three artifacts have functioning implementations and completed initial enhancements. Current efforts are focused on refinement, documentation clarity, and alignment with professional presentation standards for the final ePortfolio submission. No major technical blockers are present, but targeted instructor feedback on clarity, depth of explanation, and alignment with course outcomes would be beneficial.